DHDA-1340 BK EXP1
Black Medium Density Polyethylene Compound

Overview

DHDA-1340 BK EXP1 is a black medium density polyethylene jacket compound designed for use as a cable jacket providing excellent processing characteristics combined with reduced shrinkage, physical toughness, and superior environmental stress crack resistance (ESCR). DHDA-1340 BK EXP1 can be used by wire and cable manufacturers for the entire range of telecommunication and power applications.

Specifications

DHDA-1340 BK EXP1 meets the following material specifications

- ASTM D1248 Type II, Class C, Category 4, Grades E6, E7, E8, E9, J4, J5
- EN 50290-2-24
- Federal LP-390C Type III, Class M, Grades 2, 3, Category 4
- REA PE 39 and 89 (RAW MATERIAL SECTIONS)

Cable Specifications

Cable jackets made of DHDA-1340 BK EXP1 using sound commercial extrusion practices should meet the following specification:

- Telcordia GR-20-CORE
- ICEA-S-87-640
- ICEA S-94-649
- ICEA S-108-720
- ICEA S-97-682
- ANSI/NEMA WC74/ICEA S-93-639
- AEIC CS8
- IEC 60502, Part 2, Type ST7
- IEC 60708
- IEC 60840, Type ST7
- HD 603 S1 DMP 5, 7, 8
- DIN VDE 0818

Physical	**Nominal Value**[1]	**Unit**	**Test Method**
Density / Specific Gravity	0.945	g/cm³	ASTM D792
Melt Mass-Flow Rate (MFR) (190°C/2.16 kg)	0.80	g/10 min	ASTM D1238
Environmental Stress-Cracking Resistance (ESCR) (50 °C, 10% Igepal, F0)	> 5000	hr	ASTM D1693
Absorption coefficient, at 375 nm (abs/m)	> 400	abs/m	ASTM D3349
Hardness, Shore D (1 sec)	55		ASTM D2240
Mechanical	**Nominal Value**[1]	**Unit**	**Test Method**
Tensile Strength[2]	34	MPa	ASTM D638
Tensile Elongation[2] (Break)	900	%	ASTM D638
Thermal	**Nominal Value**[1]	**Unit**	**Test Method**
Brittleness Temperature	< -100	°C	ASTM D746
Oxidative Induction Time, 200 °C, min	100	°C	ASTM D3895

Notes

[1]Typical properties: these are not to be construed as specifications. Users should confirm results by their own tests.
[2]Measured on dogbone specimens cut from compression molded plaques

Dow DGDA-1340 BK EXP1 provides excellent surface finish, dimensional stability, and outstanding output rates over a broad range of conditions. For optimum results, use melt extrusion temperatures in the suggested range of 350 to 450°F (177 to 232°C).

Extruder Set Up	
Extruder	60-120 mm, 20:1 to 24:1 Smooth Barrel
Screw	PE Metering Screw Metering Depth: Shallow rather than deep 2:1 to 3:1 Compression Ratio
Screen Pack	20/40/60/20 Mesh
Tooling	Pressure or Tube-on tooling Draw-Down Ratio (DDR) for Tube-On: 1 to 2

Temperature Profile	**°F (ºC)**
Feed – Zone 1	360 (182)
Center – Zone 2	380 (193)
Zone 3	410 (210)
Metering – Zone 4	420 (215)
Zone 5	440 (227)
Head	450 (232)
Die	460 (238)
Melt Temperature	350-450 (177-232)

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